

ECSE 2500

Lec 2

Jan. 12

Topic : Definition of probability (problems) Axioms / Rules

□ Key Probability Terminology

Experiments, Sample space, Events, Event class/space

An **experiment** specifies a procedure and a resulting measurement. Example:

E1 : Roll a 6-sided Die and count the number of dots



E2 : Roll two 6-sided Dice and count the sum of dots

E3 : Roll two 6-sided Dice and report the dots on each die and concatenate into a 2-dimensional vector

E4 : Flip a coin 5 times and count the number of heads

E5 : Keep flipping coin and repeat until we see the first head

E6 : Flip coin 3 times and record all head/tail patterns

Continuous measurement → E7 : Pick a number at random between 0 and 1

E8 : Measure the lifetime of hard drive in terms of days/mins.

E9 : Pick a number x at random between 0 and 1, then pick another y at random between x and 1.

- Note:
- Outcomes can be discrete or continuous
 - Can be finitely or infinitely many outcomes
 - Can have same procedure but different measurements
 - Can involve a sequence of repeated experiments

□ The **sample space** is the set of all possible outcomes (measurements). Exactly one outcome from the sample space occurs on each experiment. They cannot occur simultaneously.

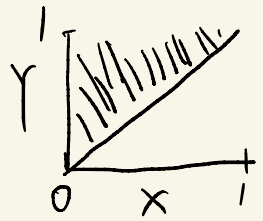
$$E1: \{1, 2, 3, 4, 5, 6\}$$

$$E2: \{2, 3, 4, 5, \dots, 12\}$$

$$E6: \left\{ \begin{array}{ccccccc} \text{HHH} & \text{HHT} & \text{HTH} & \text{HTT} & \text{THH} & & \\ \uparrow & & \uparrow & & & & \\ \text{Head} & & \text{Tail} & & & & \end{array} \right. \text{TTH, THT, TTT}$$

$$E7: \{x \mid 0 \leq x \leq 1\} \text{ or } [0, 1]$$

$$E9: \{(x, y) \mid 0 \leq x \leq y \leq 1\}$$



$$E5: \{1, 2, 3, \dots\}$$

We use notation S to denote sample space and an outcome by $\underset{\substack{\uparrow \\ (\text{delta})}}{s} \in S$.

□ An **Event** is a set of outcomes, i.e., a subset of sample space S ; $A \subseteq S$.

In the case where we have a discrete sample space and a finite # of events, we can list all possible events.

Example

$$S := \{1, 2, 3\} \text{ then}$$

what are all possible events?

= what are all subsets of sample space S ?

$$S = \{1, 2, 3\}$$

Possible events: $\left\{ \begin{array}{llll} \phi & \{1\} & \{1, 2\} & \{1, 2, 3\} \\ & \{2\} & \{2, 3\} & \\ & \{3\} & \{1, 3\} & \end{array} \right\}$

This is called the power set of S
a set of sets (a.k.a. 2^S)

Often we can specify the events in word:

A1: "The die roll was an even number" = $\{2, 4, 6\}$

A4: "We got more than 3 heads in 5 coin flips."
= $\{4, 5\}$

A8: "The device lasts between 3 to 5 years."

A9: "X and Y differ by less than 0.1."

We use the notation F to denote the event class that is the set of events we will associate probability to.

Note: - For **discrete** sample space, F is all subsets of sample space S .

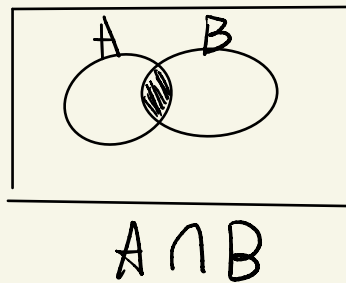
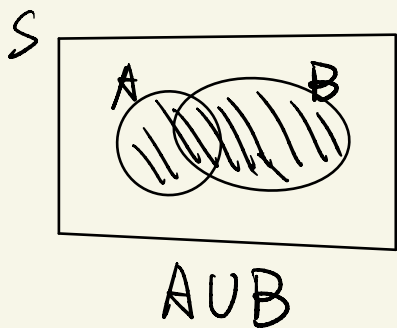
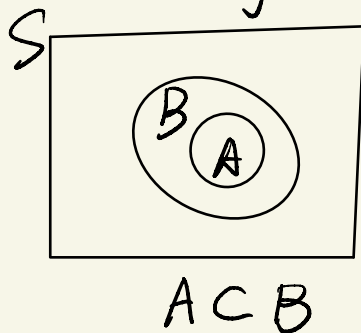
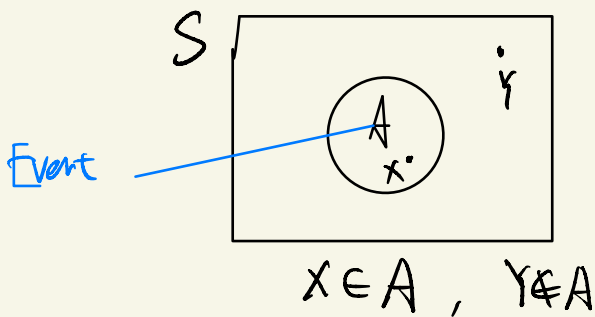
- For **continuous** sample space, F is all unions, complements and intersections of intervals such as

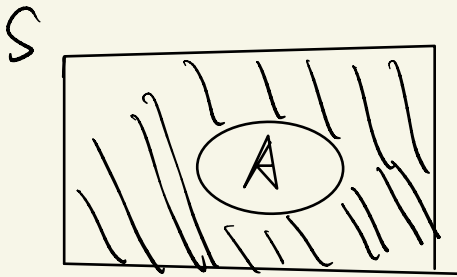
Need more complex math to rigorously define

Borel Field

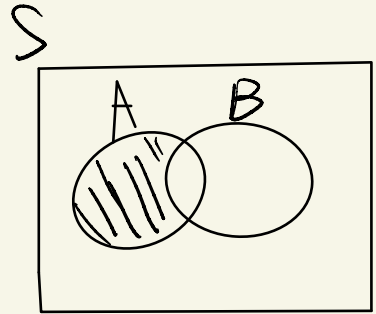
$[0, 0.5)$, $[-\infty, 12]$, $[1, 2] \cup [3, 5]$

□ A quick review of set theory

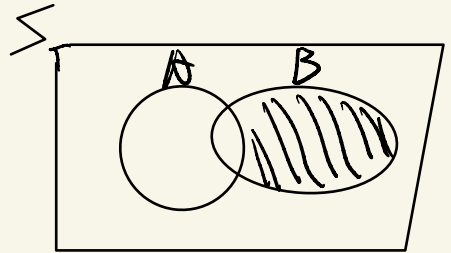




A^c



$A - B$



$B - A$